INCH-POUND

MIL-T-27/176B
4 May 2012
SUPERSEDING
MIL-T-27/176A
13 March 1991

MILITARY SPECIFICATION SHEET

TRANSFORMER, AUDIO FREQUENCY

Inactive for new design after 13 March 1991.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.

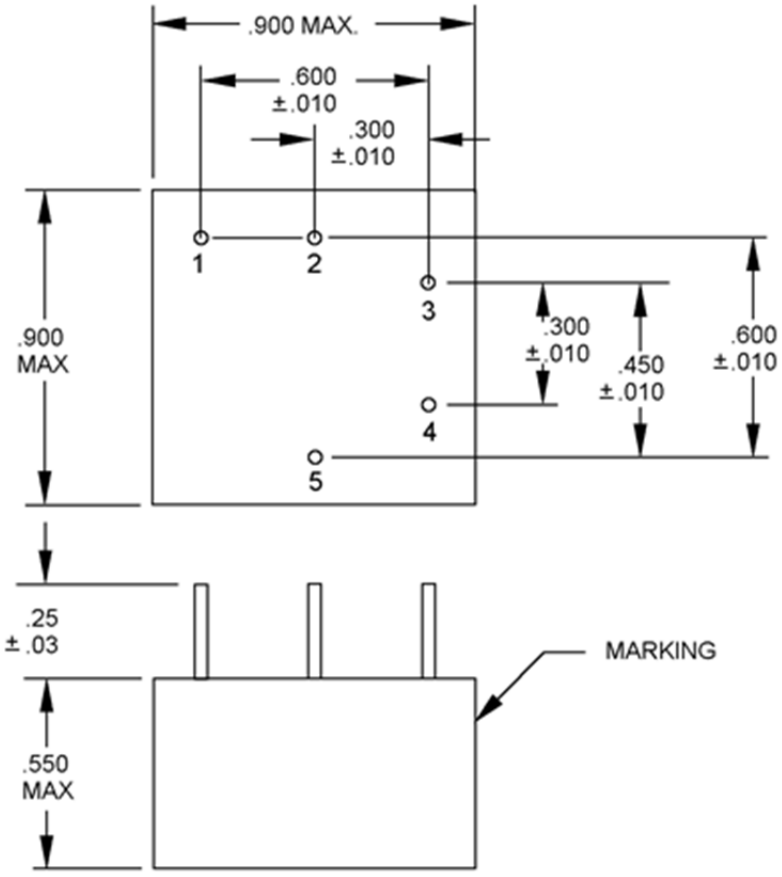
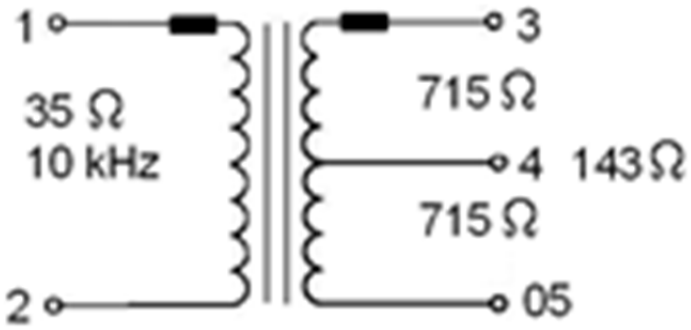


FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5950

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WORKING VOLTAGE: 100 V MAX

CIRCUIT DIAGRAM AND MARKING

Inches	mm
.010	0.25
.03	0.8
.25	6.4
.300	7.62
.450	11.43
.550	13.97
.600	15.24
.900	22.86

NOTES:
1. Dimensions are in millimeters.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configurations - continued.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings.

Source impedance (1-2): 35 ohms, 10 kilohertz.

Load impedance (3-5): 143 ohms ct.

Primary power level: 0.74 watt.

DC resistance:

(1-2): 1.15 ohms ±30 percent.
(3-4): 1.60 ohms ±30 percent.
(4-5): 1.80 ohms ±30 percent.

Working voltage: 100 volts maximum.

Design and construction:

Dimensions and configurations: See figure 1.

Material: Epoxy.

Weight: 13.61 grams maximum.

Duty cycle: Continuous.

Terminals: Pin type.

Terminal height: .25 ±.03 inch.

Operating temperature range: -55°C to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage: MIL-STD-202, method 301, test voltage 500 volts rms

Electrical characteristics:

Primary impedance (1-2): 35 ohms, 10 kilohertz.

Secondary load impedance (3-5): 143 ohms ct.

No load (center-tap voltage unbalance):

With 4 volts, 10 kilohertz, 0 ampere dc applied to (1-2).

Unbalance (3-4): ±2 percent.
Unbalance (4-5): ±2 percent.
Current in (1-2): 3.5 milliamperes maximum.
Power in (1-2): 0.014 watt maximum.
Voltage across (3-4): 4 volts ±1 percent.
Voltage across (4-5): 4 volts ±1 percent.

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Rated load:

With 35 ohms, 10 kilohertz across (1-2).

Current in (3-5): 70 milliamperes.
Voltage across (3-5): 10 volts ct.

Polarity: Additive with terminals 2 and 3 connected.

Vibration, low frequency: MIL-STD-202, method 201.

Shock (specified pulse): MIL-STD-202, method 213, test condition I.

VERIFICATION:

Qualification inspection: Not applicable.

Conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/176-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5950-2009-078)

Review Activities:
Army – AR, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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